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Visual processing is a language-agnostic window into heterogeneity in early reading development

Highlights

- Visual-processing measures identify subgroups invisible to language screeners
- Visual-challenge group are struggling readers despite good phonological abilities.
- MEP measures specifically are consistent predictors of reading risk across grades
- Unlike reading measures, visual tasks showed no differences by language or SES

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In brief

Why do some children with strong language skills still struggle to read? Ramamurthy et al. show that undetected visual-processing deficits are a key answer. By identifying hidden subgroups that language-based screening misses entirely, this work establishes a population-level empirical framework linking visual processing to heterogeneity in early reading development.

Ramamurthy et al., 2026, Current Biology *36*, 2795–2807

https://doi.org/10.1016/j.cub.2026.04.055

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Visual processing is a language-agnostic window into heterogeneity in early reading development

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https://doi.org/10.1016/j.cub.2026.04.055

SUMMARY

Reading disabilities impose profound impacts on educational and socioeconomic outcomes worldwide, yet identifying reliable early predictors remains a significant challenge. For five decades, the role of visual processing in reading development has been contested—largely due to small, homogeneous samples and a lack of research designed to capture heterogeneity in reading development. In this study, we administered theory-driven, carefully validated measures of visual processing—including multi-element processing of letters and pseudo-letters and global motion coherence—to a large, socioeconomically and linguistically diverse cohort of kindergarten and first-grade children in California public schools. A subset of children received language-based assessments and reading outcomes along with the visual measures ($n \sim 400$). Using latent profile analysis, we identified subgroups invisible to conventional screening: children with strong language skills but poor visual-processing abilities who exhibited persistent reading difficulties a year later and another subgroup of children who leveraged strong visual-processing abilities to achieve reading success despite weaker language abilities. Visual-processing measures proved equitable across children's socioeconomic status and home language, independently accounting for 12%–16% of variance in reading outcomes and emerging as consistently important predictors of reading risk in machine learning models. Integrating measures of rapid visual processing, especially multi-element processing, into early identification offers a path toward mechanistically informed personalized interventions that leverage children's specific strengths.

INTRODUCTION

Reading acquisition, a neurocognitive milestone that typically consolidates into fluent automaticity by mid-primary school, can remain an intractable challenge for children with reading disabilities like developmental dyslexia. The high prevalence of dyslexia (7%–17%[1]) across languages and cultures motivates the development of early screening tools that can be applied universally and are equally valid across language, cultural, and socioeconomic backgrounds. Even among children without a diagnosed reading disability, substantial heterogeneity exists, manifesting as marked variability in reading efficiency. This heterogeneity reflects the fundamentally multi-componential architecture of reading itself, which draws upon an intricate constellation of cognitive processes, including phonological awareness, orthographic processing, visual processing, oculomotor control, working memory, and rapid automatized naming. Understanding how different components independently and interactively contribute to variability in reading development is therefore not merely of theoretical interest—it is essential for parsing the diverse phenotypic presentations that perplex educators and clinicians and for developing both early identification tools and precision interventions tailored to individual cognitive profiles. Among these components, visual-processing mechanisms have occupied a particularly contentious position in the literature. Decades of research have linked various visual-processing deficits to dyslexia, with studies either purporting visual deficits as foundational barriers to word learning or dismissing them as epiphenomenal to core phonological deficits. Resolving this controversy requires examining not whether visual-processing matters, but rather for whom, under what circumstances, and through what mechanisms visual factors constrain or facilitate reading development. This question has practical relevance because the promise of universal screening lies not merely in identifying children who struggle but in understanding why they struggle and what unique strengths they bring to be able to develop early identification and precision intervention tools.

Theories linking visual deficits in dyslexia

Theories linking deficits in visual processing in reading disorders date back to the 1850s, when individuals struggling to read were thought to have a visual problem referred to as "congenital word blindness" in the first case study published.[2] Ever since, myriad theories have been proposed linking deficits in visual processing to reading development.[3–6] Work by Lovegrove and colleagues in the 1980s showed that people with dyslexia have challenges in the visual processing of transient stimuli.[7] Subsequent studies revealed deficits in visual sensitivity to transient and moving stimuli across a wide range of experimental conditions.[8–12] Complementary to these findings, in the 1990s, physiological evidence came from Livingstone and colleagues[13] who studied the postmortem brains of five individuals with dyslexia and compared them to five controls and reported that cell bodies in the magnocellular layers of the lateral geniculate nucleus (LGN) were fewer in terms of cell density and were approximately 27% smaller in dyslexic brains compared with control brains. Together, these observations led to the magnocellular theory of dyslexia,[4,13] which posits that difficulties in learning to read are the consequence of a low-level deficit in the magnocellular visual pathway, and that this deficit can be detected with either physiological responses[12–15] or psychophysical thresholds[9–11,16,17] to rapid, transient or moving stimuli. A number of studies that followed have failed to replicate the relationship between reading skills and motion processing.[18–22] Importantly, intervention studies revealed that intense training in reading neither improved visual motion sensitivity[10] nor did poor visual motion sensitivity hinder one's ability to show improvements in reading,[23] leaving this as the most contested hypothesis in the field. Later researchers reformulated the magnocellular hypothesis in more general terms, since the M-pathway reaches the dorsal striate cortex, implicating a more general issue with attention.[24,25] In the last decade, there has been growing interest in the visual-spatial attention hypothesis, which posits that people with dyslexia shift spatial attention more slowly than skilled readers. In particular, the exogenous or the involuntary orienting of attention has been proposed to be sluggish in individuals with dyslexia, leading to the "sluggish attention shift hypothesis," which purportedly explains various other sensory deficits reported in children with dyslexia across visual and auditory domains when processing rapidly presented information.[26–30]

Despite five decades of research, the specific visual deficits that contribute to reading development—and thus to heterogeneity in reading outcomes—remain poorly understood. The field lacks a coherent empirical framework linking visual-processing abilities to reading variability, a gap with profound implications for theory and practice.

Two fundamental challenges explain this knowledge gap. First, much of the existing literature is based on small and homogeneous samples. Most studies are not sufficiently large and diverse to discern the contribution of different risk factors, with only a couple of studies exceeding $n > 100$.[17,31,32] Secondly, the field is shifting from a "core deficit" based understanding of reading challenges to a probabilistic model that underscores reading development as more heterogeneous, shaped by multiple, interacting risk factors. Differences in visual attention, visual crowding, motion sensitivity, rapid visual encoding, and oculomotor control have all been linked to early reading outcomes, independent of language ability.[33–35] Similarly, variability in auditory processing—such as rapid auditory discrimination and temporal sensitivity—has been shown to contribute to individual differences in phonological development and reading acquisition.[36,37] These findings suggest that early reading trajectories are shaped by more than just phonological processing abilities. Capturing this heterogeneity is of great practical significance in developing precision intervention tools. Capturing heterogeneity in sensory processes, like visual-processing abilities, is particularly interesting because these abilities develop way before a child receives formal reading instructions and can therefore have tremendous potential as early identification tools.

In practice, capturing this heterogeneity, specifically early in development, requires robust, scalable tools that can be reliably administered across diverse populations early in their reading development (ages 5–6 years[3–6]). This necessitates a research design in which visual measures are administered alongside state-of-the-art language measures, with longitudinal monitoring of children's reading outcomes. The California-funded Multitudes initiative, a universal screening platform designed to identify early reading challenges in diverse populations (SB 114: Committee on Budget and Fiscal Review, Education Finance: Education Omnibus Budget Trailer Bill, 2023), provided a unique opportunity to implement a research design that would otherwise not be feasible at scale.

In this 2-year longitudinal study, a large, diverse, and representative sample of kindergarten and first-grade students in California public schools was administered conventional language-based screeners for reading difficulties. A subset of them completed two theory-driven visual measures that were carefully validated for this age group.[38] Following this, all children completed standardized reading assessments at the end of the academic school year and the following school year (2-year longitudinal study; see method details). This design provided the opportunity to investigate the importance of visual-processing abilities in predicting reading risk as early as kindergarten and to characterize children's cognitive profiles (based on performance in the language and visual tasks) to assess heterogeneity in reading development.

Visual-processing measures

Two candidate visual tasks have emerged as central to the debate on what visual-processing deficits manifest in individuals with reading challenges. The first, and the most contended, is the global motion coherence task (motion), which stems from the magnocellular theory of dyslexia. Another task that has emerged out of the attention deficit hypothesis is the multi-element processing (MEP) task (often referred to as the visual attention span [VAS] task). Onochie-Quintanilla et al.[39] provided the first longitudinal evidence that pre-reading MEP performance independently predicts future decoding skill after controlling for rapid automatic naming of objects (RAN), phonological awareness, letter knowledge, and IQ in Spanish-speaking

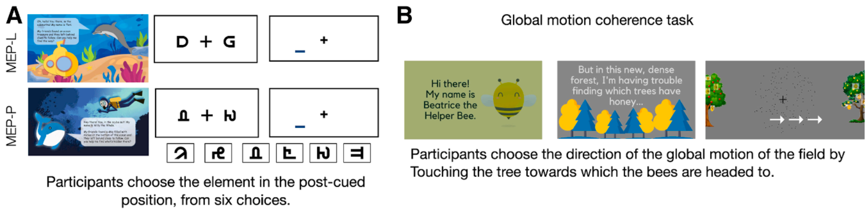

Figure 1. Visual-processing tasks schematic

(A) Schematic of the MEP task (letters and pseudo-letters version). In the multi-element processing (MEP) task, a string of letters or pseudo-letters is briefly flashed at the center of the screen, and then the participant is cued to report the identity of a single randomly chosen element. This task measures the amount of visual information that can be encoded in a brief glimpse. The design and validation of this task for reliable administration to kindergarten and first-grade children are detailed in the previous study.[38]

(B) Gamified version of the global motion coherence task modified for administration to kindergarten and first-grade children (see STAR Methods). Participants choose the direction of the dot field (gamified and presented as a colony of bees) by tapping on the apple or the lemon tree. The coherence of the number of dots carrying translational signals is modified to understand participants' motion sensitivity.

See Figure 7 for a timeline of all the measures administered in this study.

children—establishing MEP as an independent predictor of future reading. Several others have also shown that this task (1) correlates with reading ability,[40,41] (2) differs in children with dyslexia,[41,42] (3) is independent of reading experience,[43] (4) serves as a potential intervention target,[44–46] and (5) identifies a subset of poor readers with intact phonological awareness.[47] These effects are consistent across French, English, Dutch, and Chinese.[43,48] However, encoding times vary substantially across studies—for instance, Onochie-Quintanilla et al.[39] used a 4-s exposure, which the authors themselves acknowledge likely permitted serial attention shifts rather than the single-fixation capture that characterizes reading. This parametric variability across the literature makes it unclear whether comparable visual processes are being measured across studies and motivated the rigorous optimization and validation of this task specifically for kindergarten and grade 1 children reported in Ramamurthy et al.[38]

Therefore, our second candidate is the MEP task that centers on the ability to rapidly extract information from a string of visual symbols—a skill with a more direct link to reading. In this study, we administered the MEP task that was rigorously optimized for reliability in kindergarten and first-grade children (for a detailed design and validation of the MEP measures, see Ramamurthy et al.[38]) and the motion task adapted from a previous study,[31] tailored for administration to kindergarten and first-grade children (Figure 1). While both tasks are widely studied, their utility in identifying reading-related risk remains unclear, especially in large, diverse, representative samples.

Our goal in this study was to investigate whether developmentally appropriate, scalable visual-processing measures—used alongside state-of-the-art language-based screening tasks—can predict reading difficulties independent of environmental factors like the child's home or instructional language.[15–19]

RESULTS

LPA reveals subgroups of children with specific visual strengths and specific visual challenges

Children beginning formal reading instruction exhibit marked heterogeneity in their reading development, yet current screening approaches—focused predominantly on language-based measures—may obscure meaningful cognitive profiles. To investigate heterogeneity in reading outcomes, we implemented Bayesian information criterion (BIC)-optimized latent profile analysis (LPA) via the *Mclust* package in R,[49] on the kindergarten data (n = 157). This revealed five latent profiles as the optimal solution (Figure S5). Subgroup criteria were derived from the LPA solution, based on the pattern of performance exhibited by each subgroup, on kindergarten children. Subsequently, this criterion was applied to define profiles consistently across children from both grades (see STAR Methods).

Figure 2A shows the profiles of strengths and challenges in each subgroup. Three of the subgroups represented expected patterns of performance: (1) an above-average-performing group (n = 26) that performed more than 0.25 standard deviations above the mean on all tasks, (2) an average-performing group that performed around the mean (n = 62), and (3) a below-average-performing group that performed more than 0.25 standard deviations below the mean (n = 30). Additionally, two interesting subgroups emerged: children with above-average performance in MEP tasks and lower abilities in other domains—the specific visual strength group (n = 16, 10% of total kindergarten children)—and children with low performance on MEP tasks who exhibited high performance across other domains—the specific visual-challenge group (n = 23, 15% of total kindergarten children). Notably, the reading outcomes a year later for children in each of these subgroups presented a pattern where the specific visual strength group showed longitudinal reading outcomes comparable to the above-average-performing group, indicating that strengths with MEP can be a protective factor for reading development. The specific visual-challenge group, on the other hand, showed longitudinal reading outcomes close to the average-performing group, even though they had strong language skills (see Figure 2B).

Figure 2C shows the reading outcome (Woodcock-Johnson Letter-Word Identification [WJ-LWI]) trajectories across time—from the concurrent year to the 1-year follow up. A linear mixed-effects model (specified as reading outcome $\sim$ time $\times$ subgroups + (1|student)) revealed a significant main effect of time, $F(1, 152) = 23.420$, $p = 3.169 \times 10^{-6}$, indicating overall improvement in reading outcome across the year. There was also a significant main effect of subgroups, $F(4, 152) = 4.88$, $p = 9.922 \times 10^{-4}$, with above-average performers and specific visual strength groups showing significantly higher baseline scores than below-average performers (p = 0.000489 and p = 0.0418, respectively). Although the time $\times$ subgroup interaction was not significant ($F(4, 152) = 1.765$; $p = 0.139$), estimated marginal means revealed that all subgroups—except below-average performers—exhibited statistically significant reading gains over time.

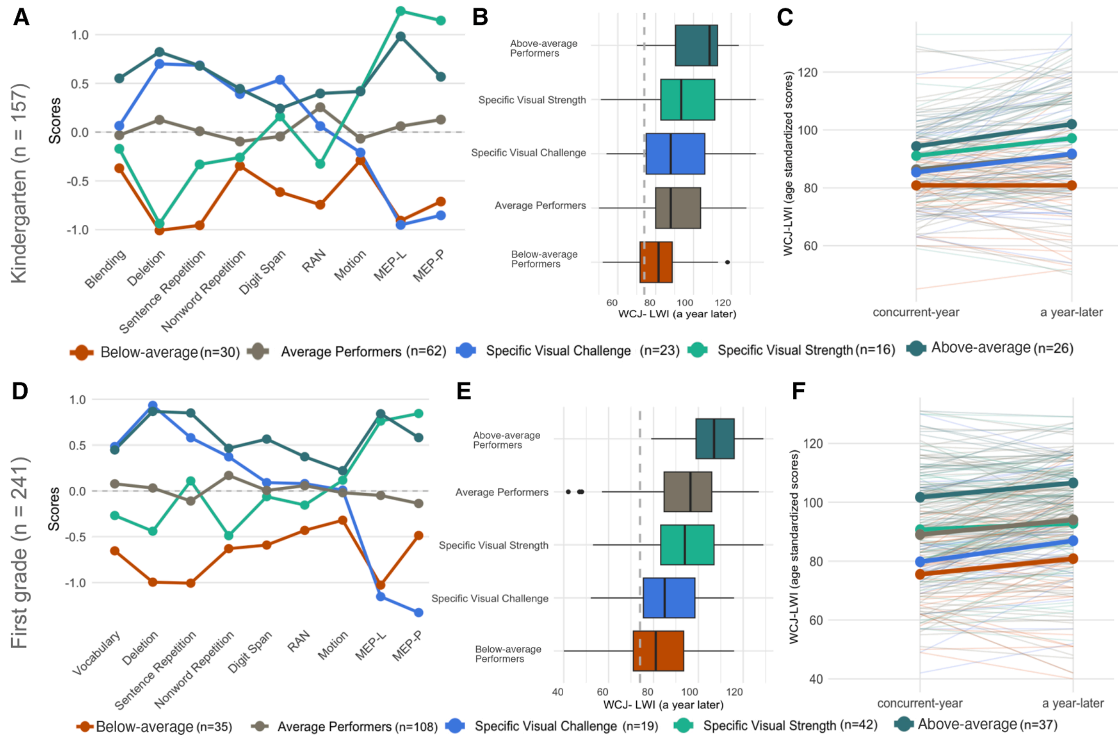

Figure 2. Performance profiles derived from LPA

(A–C) Kindergarten. Latent profile analysis (LPA) identifies cognitive subgroups with distinct reading trajectories in kindergarten children. (A) Performance profiles derived from LPA based on *Z* scored performance across visual-processing and language measures. (B) Distribution of Woodcock-Johnson Letter-Word Identification (WJ-LWI) scores 1 year later, showing stratified reading outcomes across subgroups. (C) Individual reading growth trajectories from baseline to follow up, with bold lines indicating group-level means.

(D–F) First grade. Performance-based subgroups and reading outcomes in first-grade children ($n = 241$) mirror kindergarten profiles with developmental shifts. (D) *Z* scored performance profiles derived from the performance criteria applied from the kindergarten group. (E) Distribution of WJ-LWI scores at follow up 1 year later, showing continued stratification in reading ability. The dotted gray line indicates the 20th percentile cutoff on WJ-LWI scores. Typically, children with scores below this cutoff are classified as at risk. (F) Longitudinal reading trajectories for each child, with group-level means shown in bold. Boxplots disply median (center line) interquartlie range (IQR) (box) and 1.5× IQR (whiskers), with outliers shown as individual points.

See also Figures S5 and S6.

Importantly, initial disparities in reading performance persisted, with below-average performers showing no measurable growth.

The same clustering criteria were applied to the first-grade cohort ($n = 241$) (see Figure 2D). The specific visual-challenge group continued to rank closer to the low-performing subgroup, and the specific visual strength subgroup was close to the average-performing subgroup in reading outcomes a year later (see Figure 2E). Figure 2F shows the reading outcome (WJ-LWI) trajectories across time—from the concurrent year to the year after. A linear mixed model (specified as reading outcome ~ time * subgroups + (1|student)) revealed a significant main effect of subgroups, $F(4, 236) = 10.972$, $p = 3.544 \times 10^{-8}$, like kindergarten, below-average performers < specific visual strength group < above-average performers. The interaction between time and subgroups was not significant, $F(4, 236) = 0.275$, $p = 0.894$, suggesting that although all groups differed, their rates of growth were comparable. Notably, the specific visual strength group maintained reading levels on par with average performers, reinforcing their more resilient developmental trajectory.

These results show that children with strong language but poor visual-processing abilities (the specific visual-challenge group) exhibit persistent risk for lower reading outcomes. Conventional risk classifications typically use reading scores below the 20th percentile to classify children as at risk, and Figure 3 shows the percentage of children in each subgroup that are below this conventional risk criterion. Specific visual-challenge subgroups consistently show a higher percentage of children at risk after the below-average-performing subgroup across both grades (Figures 3A, in kindergarten, and 3B, first-grade children, respectively).

The specific visual-challenge subgroup exemplifies the clinical significance of undiagnosed visual-processing issues. These children—23 in kindergarten and 19 in first grade—demonstrated average to above-average performance on all language-based measures yet exhibited poor reading outcomes. Conventional screening would miss these children entirely, as their phonological processing abilities fall within or above the normal range. This would perplex educators and clinicians, as these children are unlikely to benefit from phonological-focused interventions, as their phonological skills are already adequate. Instead, their reading difficulties stem from visual-processing deficits that remain undetected without targeted assessment.

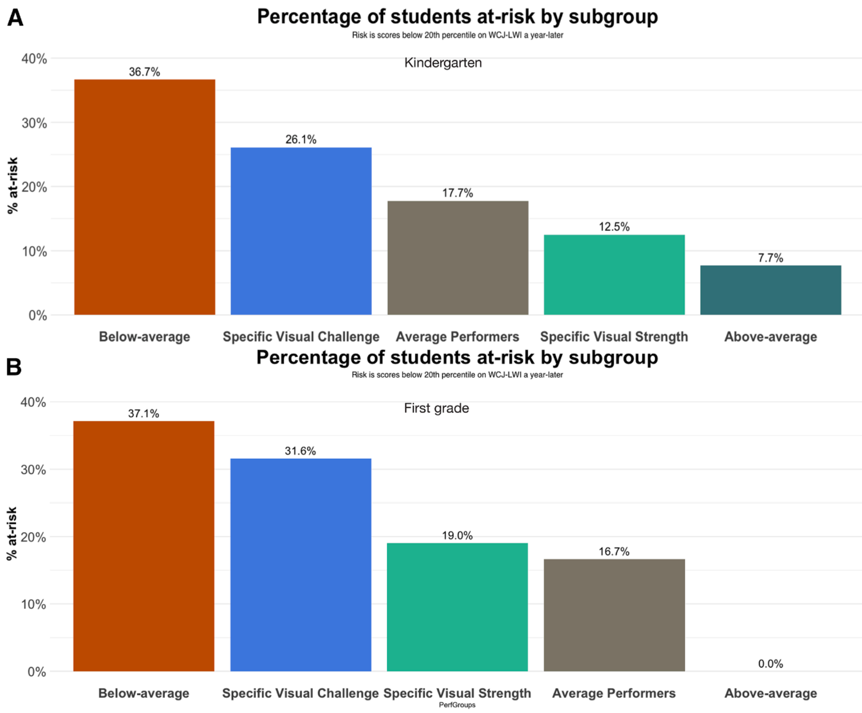

Figure 3. Risk distribution within each subgroup

(A) Shows the percentage of children at risk in kindergarten based on conventional classification that oftentimes classifies a child as at risk for reading challenges if standardized reading scores are below a certain cutoff. We used a cutoff of 20th percentile on standardized reading outcome measures based on a previous study (see STAR Methods). We used WJ-LWI scores a year later for risk classification. Representations here corroborate with Figure 2B presented above. The below-average-performing subgroup had the highest percentage of children at risk, followed by children in the specific visual-challenge group with the least percentage of children in the above-average-performing subgroup.

(B) Shows the percentage of children at risk based on a cutoff score of below the 20th percentile on WJ-LWI scores obtained 1 year later. Representations here corroborate with Figure 2E presented above for first-grade children. Similar to kindergarten, the below-average-performing subgroup had the highest percentage of children at risk, followed by children in the specific visual-challenge group.

See also Figure S6.

This subgroup underscores the necessity of incorporating visual-processing measures into early screening protocols.

By contrast, children with strong visual-processing abilities demonstrate comparable reading performance to above-average performers, suggesting that there might be compensatory mechanisms early in development that those with specific visual strengths can leverage to build reading skills. Understanding the mechanisms that drive these differences can pave the way for precision intervention strategies.

Visual measures are important predictors of reading risk

The identification of distinct performance subgroups with substantially different risk profiles—ranging from 0% at risk in above-average performers to 37% in low performers—raises a critical question: can individual reading risk be predicted from baseline language and visual-processing measures? Two important considerations guide our decision-making: (1) reading difficulties are not primarily visual-processing disorders, although visual deficits may co-occur and hold predictive value, and (2) robust evaluation of early visual markers requires a multi-year longitudinal design and a sufficiently large sample. The sample sizes of children with all measures varied across years. Concurrent reading outcome measures (Spring 2023) were available for 108 kindergarteners and 216 first-grade children. Longitudinal follow up (Spring 2024) was available for K: 53, G1: 112 (see Figure 7 for timeline of task administration). We performed multiple imputations to address missing data within language, visual, and reading outcome domains before merging imputed datasets. The final sample size, after imputations, was K: 157 and G1: 241 (see STAR Methods). We defined risk as any child with a WJ-LWI score < 20th percentile being classified as at risk, based on a previous study.[50]

In our sample, 20% of children were identified as at risk (kindergarten: 32/157; grade 1: 51/241), resulting in a notable class imbalance. Class imbalance is an expected feature when the condition of interest (reading risk) has a relatively low base rate. Traditional classification metrics like sensitivity and specificity are sensitive to class distribution and therefore can be misleading with the current sample size. As we continue to add cohorts, the absolute number of at-risk children will increase, allowing for more robust estimation of these metrics. We fit a risk prediction model with leave-one-out cross validation (LOO-CV) on all available reading outcome measures. The goal was to identify the minimum number of predictors that maximized the area under the curve (AUC) and accuracy (see Table S4). AUC provides a threshold-independent measure of model performance that is particularly well-suited to imbalanced datasets like ours. Table S4 shows the minimum number of predictors that maximize AUC and accuracy for each reading composite outcome that is available for each grade, respectively. Risk status was computed as below the 20th percentile for each reading measure (see STAR Methods for a detailed description of each reading measure and composite scores available). The AUC-accuracy profiles presented in Table S4 reinforce the role of MEP-L as one of the key predictors. In the absence of MEP-L, MEP-P is one of the stable predictors that maximizes AUC and model accuracy (see Table S5), likely due to its high disattenuated correlation ($r = 0.91$[38]) with the MEP-L measure.

Next, we employed a random forest classification to identify children at risk for reading difficulties (defined as scoring below the 20th percentile on standardized reading outcome WJ-LWI in our dataset) and to determine which specific measures (abilities) were most predictive of reading outcomes. Given the imbalance in risk prevalence, we focus on feature importance metrics to determine the predictive value of the visual-processing measures.

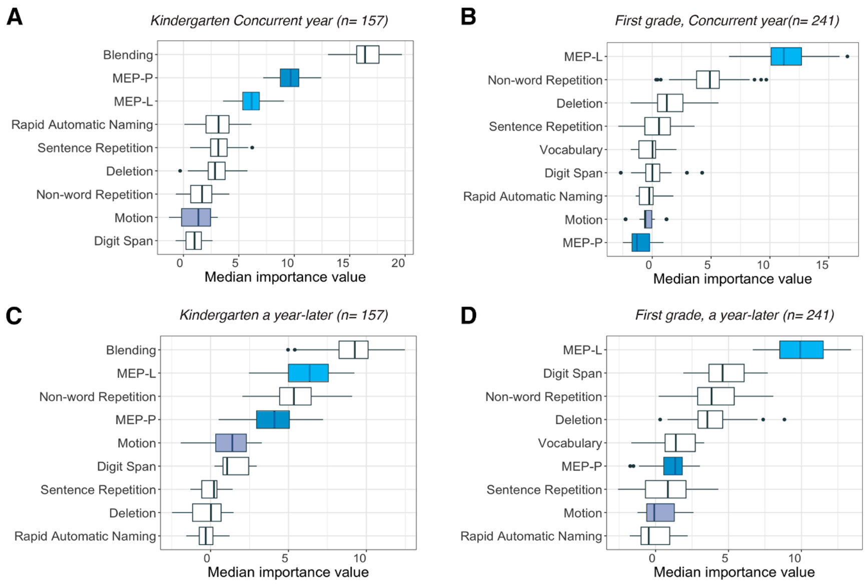

Figure 4. Feature importance plots for predicting reading risk

(A and B) Show feature importance plots for predicting risk at the end of the same year across both grades.

(C and D) Show feature importance plots for predicting risk in children who were longitudinally followed up a year later.. See also Figure S4, which shows the exact plot for simulated class-balanced data. Across both the class-balanced and imbalanced datasets, the MEP measures are important predictors of risk alongside other language-based measures. Boxplots display the median (center line), interquartile range (box), 1.5× IQR (whiskers), with outliers shown as individual points. See also Tables S4 and S5.

To interpret how individual measures contributed to the predictions, we computed feature importance values using the Boruta package in R.[51] In Figure 4, we present feature importance plots to quantify the relative importance of each predictor—such as MEP, deletion, blending, etc.—on the model's ability to classify children at risk. We present risk classification based on the reading outcomes measured in the concurrent school year (Figures 4A and 4B) and those obtained a year later (Figures 4C and 4D) for each grade, respectively.

Higher median importance values indicate that the measure was frequently used to split the decision nodes across the different decision trees in the random forest and thereby consistently improves prediction accuracy when included in the prediction model. Our results revealed visual measures, particularly MEP, as consistently important predictors across both time points and grades. It should be noted that since we have fewer children at risk, models may prioritize variables that simply distinguish the majority-class patterns, thereby skewing feature importance estimates. To verify that our findings reflected true patterns rather than imbalance artifacts, we replicated our analysis with class-balanced datasets using the ROSE package in R[52] that simulated a balanced dataset by synthetically oversampling minority-class instances (at risk) and undersampling the majority class. Figure S4 is identical to Figure 4 but plotted with the simulated balanced dataset. We found that the visual measures, especially MEPs, emerged as an important and stable predictor of risk in kindergarten, and by first grade, only MEP-L was a significantly important predictor. This pattern held true for risk predicted at the end of the school year as well as at the 1-year follow up.

Visual-processing measures show equitable performance across diverse learners while predicting reading outcomes

The preceding analyses demonstrate that visual-processing measures reveal meaningful heterogeneity in reading development and are important predictors of reading risk. However, before concluding that visual-processing measures offer added value for screening, we must empirically look for biases in visual measures. Our motivation to investigate this question stems from the fact that socioeconomic status impacts educational opportunity and academic outcomes.[53–55] Measures of letter knowledge, phonological awareness, and other language skills often reflect the unequal opportunities afforded to children due to socioeconomic factors.[56,57] Oftentimes, screening measures reflect disparities in a child's experience prior to formal schooling and are not informative in terms of potential neurobiological intervention targets.[58,59] A child's financial and social resources are among the strongest predictors of performance on reading assessments, and the socioeconomic disparity in elementary school reading scores has grown by over 40% in the second half of the 20th century.[54] In our own dataset, we observed significant group differences in the reading outcomes when children were grouped based on free and reduced-price meal (FRPM) eligibility (free or reduced-price meals, a metric highly correlated with county poverty rates 0.91–0.95[60] and a student-level indicator of home income) and reported primary home language (see Figures 5D–5F and 6D–6F). Such bias undermines the validity of many screeners: if screening measures are confounded with a child's eligibility for FRPM or the child's home language, the observed correlations with reading outcomes could reflect shared environmental factors rather than any underlying mechanism. We therefore examined whether visual task performance differs when children are grouped by (1) their eligibility for FRPM and (2) reported primary home language.

We found no group differences in performance on the visual tasks across children grouped based on FRPM status (Figures 5A–5C) or primary language (Figures 6A–6C). We performed a t-test to compare across groups and adjusted the significance threshold for multiple comparisons (Bonferroni-adjusted p threshold: 0.004 (0.05 (p-cutoff)/12 comparisons)). Figures 5A–5C show how children from the FRPM-eligible and -ineligible groups perform across all visual tasks, and Figures 6A–6C show how children from the English-speaking and Spanish-speaking groups perform across all visual tasks. Children as young as kindergarten exhibited no group differences in task performance across all visual-processing measures. This is in stark contrast to the reading outcome measure (WJ-LWI) in our dataset, which showed group

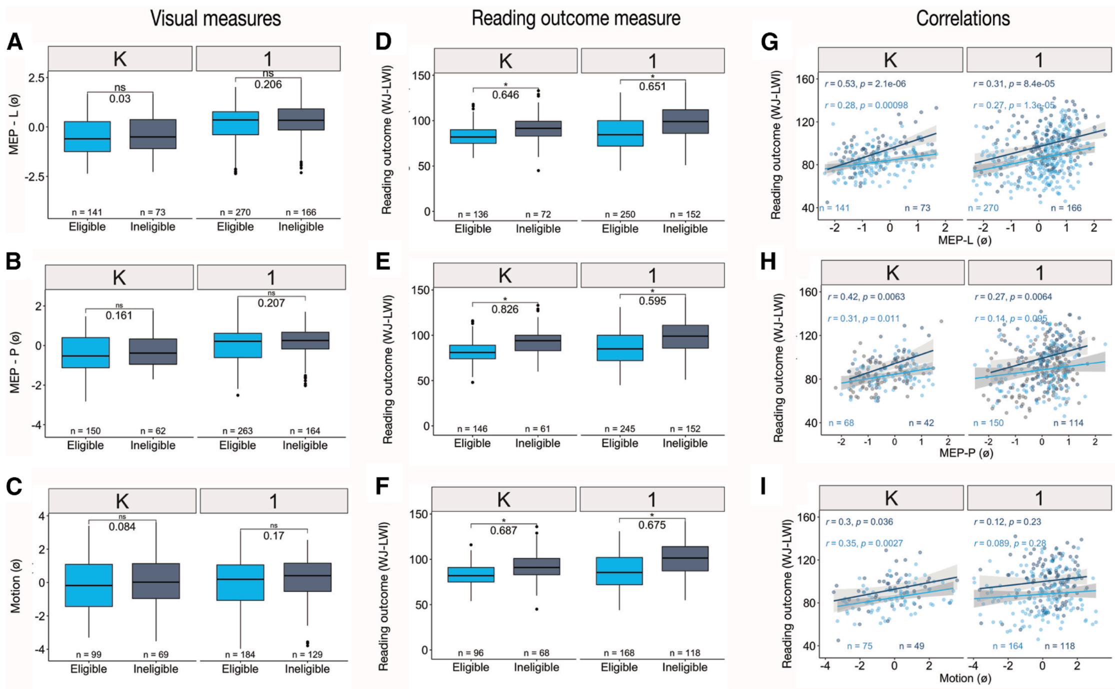

Figure 5. Performances based on FRPM eligibility groups

(A–C) The first column shows differences in task performance in the MEP-L, MEP-P, and motion (MEP-L, multi-element processing of letters; MEP-P, multi-element processing of pseudo-letters; motion, global motion coherence discrimination) measures across the FRPM-eligible and -ineligible groups for both grades.

(D–F) Shows differences in reading outcome measure (Woodcock-Johnson Letter-Word Identification [WJ-LWI]) for the same participants who had a reading outcome measure. The number between the boxplots is the effect size (Cohen's d) between the groups, and asterisks show those groups that are significantly different (*$p < 0.004$). Boxplots display the median (center line), interquartile range (box), and 1.5× IQR (whiskers), with outliers shown as individual points.

(G–I) Shows the correlation between visual measures and reading outcome for each grade, respectively. The colors correspond to the group split based on FRPM presented in (A)–(F), respectively.

See also Figures S1, S4, and S7 and Tables S1 and S2.

differences as early as kindergarten. First-grade children showed a significant group difference only in the letter version of the MEP task. Since the MEP-L task involves letter knowledge, it is possible that factors like preschool attendance, home environment, and parental education could mediate the group difference we notice between the English and Spanish proficient groups by first grade. No such group difference was noticed in the pseudo-letter version of the task.

These findings suggest that the visual measures capture performance that is largely independent of socioeconomic status and home language—in contrast to reading outcomes, which showed group differences as early as kindergarten. This supports the interpretation that correlations between MEP performance and reading outcomes are not influenced by shared socioeconomic or linguistic factors (see Figures 5G–I and 6G–6I for correlations).

Based on $n = 539$ students with all the visual measures and end-of-the-year reading outcome measures, we investigated the relationship between visual processing, reading outcome in kindergarten ($n = 108$) and first grade ($n = 240$) students, and SES defined by FRPM eligibility. Across both grades, we fit two linear regression models: (1) a visual-only model (we included a model with just the non-alphabetic visual tasks presented in Table S2) and (2) a visual with FRPM eligibility interactions as shown in the equations summarized in the STAR Methods section, and correlations are presented in Figures 5G–5I.

First, the model including MEP-L, MEP-P, and motion as predictors explained 18.22% of the variance in reading outcome in kindergarten and 9.42% in first grade (see Table S1). Only MEP-L performance was a significant predictor for kindergarten (β = 0.218, $p = 0.0253$) and first grade (β = 0.307, $p = 0.00063$). This is because performance in the MEP-L and MEP-P tasks is highly correlated ($r = 0.7$) and correlated with the motion task ($r = 0.3$); however, each measure is a significant univariate predictor of reading outcome across both grades (see Table S2). An ANOVA comparison between the visual-only model and the model with visual and FRPM interactions demonstrated that including the FRPM eligibility improved model fit for both kindergarten ($F(4,100) = 3.930$, $p = 0.00526$) and first graders ($F(4, 232) = 5.396$, $p = 0.00035$). The lack of significant interactions indicates that the relationship between visual measures and reading outcomes does not vary based on FRPM eligibility (as noted in Figures 5G–5I). While children with different FRPM eligibility did not differ significantly in their visual-processing abilities, FRPM eligibility emerged as a strong predictor of reading outcome as expected.

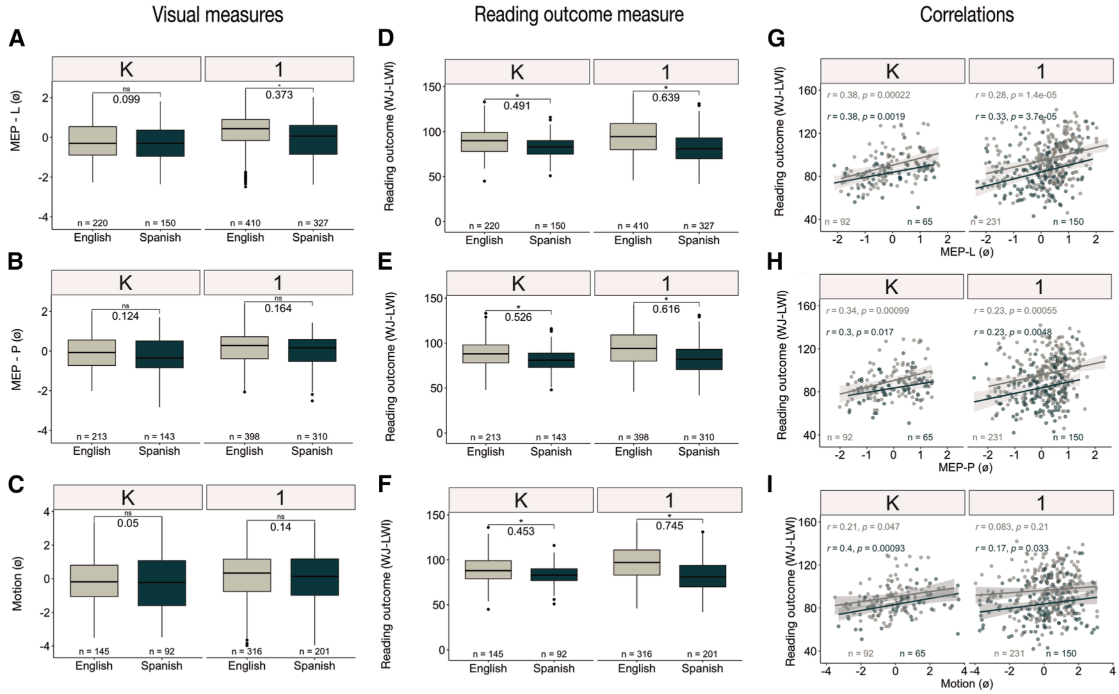

Figure 6. Performances based on primary oral language groups

(A-C) The first column shows differences in task performance in the MEP-L, MEP-P, and motion (MEP-L, multi-element processing of letters; MEP-P, multi-element processing of pseudo-letters; motion, global motion coherence discrimination) measures across the two primary language groups for both grades.

(D–F) The second column shows the differences in reading outcome measures (Woodcock-Johnson Letter-Word Identification [WJ-LWI]) for the same participants who also have reading outcomes. The number between the boxplots is the effect size (Cohen's d) between the groups, and asterisks show those groups that are significantly different (*$p < 0.004$). Boxplots display median (center line), interquartile range (box), and 1.5× IQR (whiskers) with outliers as individual points.

(G–I) Shows the correlation between visual measures and reading outcome for each grade, respectively. The ivory and green correspond to the group split based on primary language presented in (A)–(F), respectively.

See also Figure S2 and Tables S2 and S3.

We next investigated the relationship between visual-processing measures and reading outcome across different primary language groups in kindergarten (n = 157) and first grade (n = 381). We fit two linear regression models: (1) a visual-only model (we included a model with just the non-alphabetic visual tasks presented in Table S2) and (2) a visual with primary language interactions (correlations are presented in Figures 6G–6I and Table S3). The visual-only model explained 16.3% of the variance in the reading outcome measure in kindergarten and 11.9% in first grade. Only MEP-L performance was a significant predictor for kindergarten ($\beta = 0.212$, $p = 0.0107$) and first grade ($\beta = 0.316$, $p = 2.51 \times 10^{-6}$). An ANOVA comparison between the visual-only model and the model with visual and primary language interactions demonstrated that including the primary language improved model fit for both kindergarten ($F(4,149) = 2.993$, $p = 0.021$) and first graders ($F(4, 373) = 7.896$, $p = 4.058 \times 10^{-6}$). The lack of significant interactions indicates that the relationship between visual-processing measures and reading outcomes does not vary between English and Spanish speakers. Primary language, however, emerged as a strong predictor of reading outcome measures. Overall, visual-processing measures accounted for ~16%–18% of the variance in reading outcome in kindergarten and ~10%–12% in first grade.

DISCUSSION

Despite five decades of controversy, the debate over whether individual differences in visual processing can explain individual differences in reading ability remains active.[3–5,7,34] This might seem surprising in light of hundreds of studies that report visual-processing deficits in children with dyslexia as well as correlations to reading scores.[31,33,34,37,61,62] Progress has been hindered by small, homogeneous samples constrained by lab-based research designs. Despite best efforts to recruit representative samples, the families who respond to university recruitment efforts are rarely representative of regional or national diversity, yet it is precisely this heterogeneity that must be understood to develop appropriate screening tools and intervention strategies. This dual constraint—methodological and conceptual—has kept visual processing at the periphery of reading science, despite long-standing debate of its relevance in dyslexia and reading challenges. Central to this constraint is the long-standing unresolved understanding about how visual-processing abilities might contribute to heterogeneity in reading development. Here, we were able to address these limitations due to the opportunity to sample directly through school partnerships. This was possible due to the extensive outreach by the state-funded University of California, San Francisco's (UCSF's)

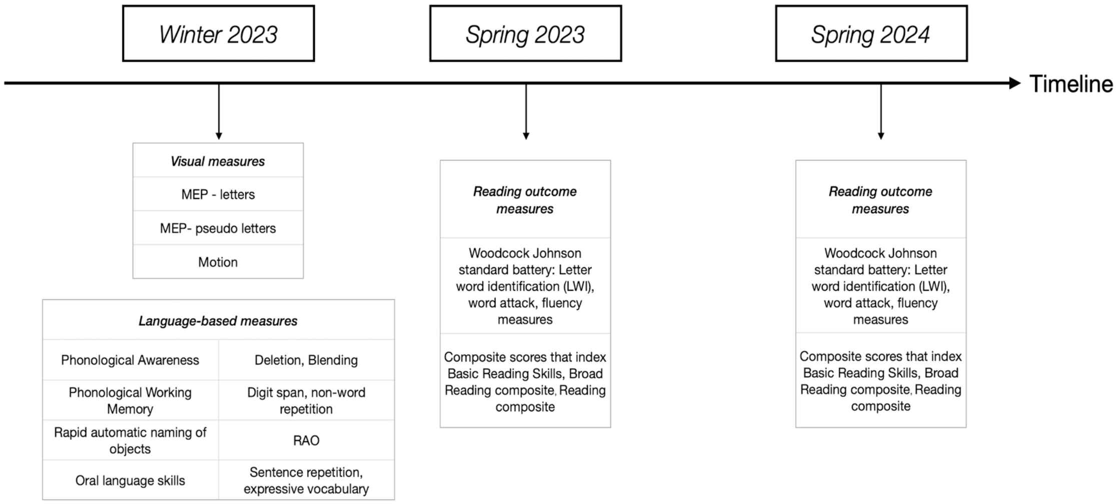

Figure 7. Timeline of the administration of various measures

Note that the visual and language measures were administered in the winter of 2023 and two time points of the reading outcome measures in spring 2023 and spring 2024.

See also Figure S3 for excluded data.

Multitudes initiative to develop demographically age-appropriate screening tools for early identification of dyslexia.

Although a platform that enables large outreach is necessary, it is not sufficient until we have meaningful measures. In a literature that has at least five different main theories and many paradigms with often contradictory findings, it was important to select visual-processing measures based on prior evidence of their ability to identify struggling readers missed by conventional language-based assessments. The MEP-L and MEP-P tasks used here were developed and validated through a systematic multi-study process described in our previous work. Briefly, the task was iteratively modified for K/1 children—adjusting encoding time and string length—and optimized using IRT-based item selection, yielding a ~5-min measure with empirical reliability of 0.86 that reliably predicts end-of-year reading outcomes. This was a critical first step for use with young children in a school setting (see Ramamurthy et al.[38] for information on the demographics of this sample).

Global motion discrimination tasks, often invoked to probe magnocellular (M-pathway) integrity, have yielded mixed evidence for a direct mechanistic role in reading acquisition.[63] By contrast, MEP tasks are ecologically closer to the demands of word reading: they index how efficiently a child can encode multiple visual elements within a single fixation—precisely the perceptual bottleneck that fluent reading must overcome. It is therefore unsurprising that MEP outperforms the motion task in forecasting later reading outcomes. Critically, this pattern is reinforced by our LPA: the two unique subgroups—the specific visual strength and specific visual-challenge groups—were characterized by dissociations in MEP performance while exhibiting relatively homogeneous motion discrimination abilities (Figures 2A and 2D), providing convergent support for the view that rapid, fixation-bound visual encoding—not low-level motion processing—is the more critical constraint on early reading development.

A further limitation concerns the interpretation of the MEP task. Although the 240 ms encoding window is specifically designed to capture fixation-bound visual encoding—consistent with the Sperling partial report paradigm—tasks involving rapid MEP also place inherent demands on visuospatial working memory. In this dataset, MEP performance was largely independent of digit span, a verbal working memory proxy, in kindergarten (MEP-L: r = 0.107, p = 0.184; MEP-P: r = 0.130, p = 0.104) and showed only small correlations in grade 1 (MEP-L: r = 0.253; MEP-P: r = 0.202), suggesting the observed associations with reading are not primarily driven by verbal working memory capacity. However, we did not have a direct measure of visuospatial working memory, such as the Corsi block task (see Figure 7 for all the measures administered in this study).[64] Future work should include such measures to more precisely disentangle the contributions of rapid visual encoding and visuospatial working memory to MEP task performance and its relationship with early reading development.

Our random forest classification and AUC-based analyses further demonstrated that particularly MEP measures are consistent predictors of reading risk (calculated using reading scores from the concurrent year and a year later). The importance of MEP as a predictor aligned directly with the profiles identified through LPA. This convergence between predictive importance and heterogeneity suggests that MEP captures processes relevant to reading development that are orthogonal to language abilities. While our initial aim was to directly compare visual-only versus language-only prediction models, such comparisons require substantially larger samples with sufficient representation of at-risk children—a direction for future work as our longitudinal cohort expands.

Performance on visual-processing tasks showed no differences by home language or socioeconomic status (FRPM eligibility). A critical question, however, is whether the latent profiles themselves are evenly distributed across demographic groups or whether subgroup membership interacts with these factors in ways that reveal hidden compositional biases. For example, are children in the below-average-performing subgroup disproportionately composed of Spanish-speaking or FRPM-eligible children, suggesting that subgroup classification inadvertently recapitulates existing disparities? We examined subgroup composition by FRPM eligibility (Figures S6A and S6B) and primary home language (Figures S6C and S6D). We emphasize that the sample sizes within individual subgroups are small, precluding strong statistical inference, and we present these findings as hypothesis-generating for future investigation with adequately powered samples. First, based on FRPM, we did not observe marked distinctions across most subgroups. By contrast, subgroup composition based on home language revealed differences. Above-average performers were predominantly English-speaking, whereas below-average performers were overwhelmingly Spanish-speaking (across both grades) and ranked high and low on the reading outcomes, reflecting that the reading outcome measures were all in English. Average performers maintained a balanced language composition. With respect to home language, in kindergarten, the specific visual strength group largely consisted of Spanish-speaking children. And the specific visual-challenge group at kindergarten primarily consisted of English-speaking children. Conversely, in grade 1, the specific visual strength group was predominantly English-speaking children, and the specific visual-challenge group flipped to being predominantly Spanish-speaking similar in composition to the below-average-performing group, reversing the pattern observed in kindergarten (see Figure S6). These patterns raise critical questions about how these different subtypes of reading challenges might contribute to the specific difficulties faced by these children and the kinds of early interventions that might mitigate the impact on future reading outcomes.

This study provides compelling empirical evidence that differences in rapid visual processing contribute meaningfully to heterogeneity in early reading development. Without the visual measures, the specific visual-challenge group would have mimicked the above-average performers in terms of their scores in the language-based tasks, making it hard to reconcile their poor reading outcomes a year later. By contrast, children in the specific visual strength subgroup would have mimicked the below-average performers in terms of their scores in the language-based tasks. Their particularly high performance in the MEP tasks seems to have played a compensatory role based on their reading outcomes that was comparable to the above-average performers. These findings have profound implications for how we design, implement, and interpret early screening programs.

As these visual measures have now been adopted into two of the four universal screeners recommended by the State of California,[50,65–71] forthcoming cohorts will offer valuable opportunities to evaluate the predictive utility—particularly the sensitivity and specificity—of various combinations of predictors over longer timescales. A central limitation of this study is that the language-based measures administered were not originally designed with the linguistic and demographic diversity of California public schools in mind. This may have attenuated estimates of language skill contributions and limits the generalizability of findings based on these measures alone. Relatedly, missing data was high for motion (34% kindergarten; 27% grade 1) and several language measures; although a sensitivity analysis confirmed the latent profile structure was robust to imputation on available complete cases, findings involving these specific measures should be interpreted cautiously. As part of the state-funded Multitudes initiative, language assessments have since been redesigned and calibrated for California's diverse learner population, complementing the equitable visual measures introduced here. Thus, future cohorts will help replicate and extend the profile-based classification framework developed here with larger and more complete samples.

Our findings challenge long-standing assumptions about the role of vision in reading and establish a powerful framework to reconceptualize the mechanisms driving heterogeneity in reading development. By establishing a population-level framework for longitudinal follow up, this work enables future studies to trace how visual and language-based risk factors interact over time. In doing so, this research lays the groundwork for a new generation of early identification tools—ones that are inclusive, developmentally appropriate, and empirically grounded—opening the door to precision interventions that can transform outcomes for young readers worldwide.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Mahalakshmi Ramamurthy (maha10@stanford.edu).

Materials availability

This study did not generate any new, unique reagents.

Data and code availability

- The experiment code for the MEP tasks is available at https://github.com/yeatmanlab/roar-mep and deposited at Zenodo https://doi.org/10.5281/zenodo.19429971. The experiment code for the motion processing task is available at https://github.com/yeatmanlab/roar-mp and deposited at Zenodo https://doi.org/10.5281/zenodo.19425956.
- De-identified data are available in the data repository (https://osf.io/85h6e/files/osfstorage).
- The analysis code is available at https://github.com/yeatmanlab/Visual-Tasks-as-Language-agnostic-Early-Identification-Measures-of-Reading-Challenges and deposited at Zenodo https://doi.org/10.5281/zenodo.19429953.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

ACKNOWLEDGMENTS

This work was supported by funding provided by the State of California through Senate Bills 109 and 129 and Assembly Bill 103 of 2019, 2021, and 2022, respectively, in item 6440-001-0001; the University of California, San Francisco; the Charles and Helen Schwab Foundation; the National Institute on Deafness and Other Communication Disorders (K24C015544); the National Institute of Neurological Disorders and Stroke (RF1NS050915); the Stanford Maternal Child Health Research Institute; R01HD095861; and Stanford Impact Labs. We want to thank the Multitudes central operation team: Meredith Kalmes and Jessica Clyne for school scheduling and staff coordination and Heather Elgin for training the proctors. We want to also thank the incredible proctor team that collected data across the State of California and the teachers and administrators who made this possible.

AUTHOR CONTRIBUTIONS

M.R. conceived the study, developed the methods, performed all formal analysis and data curation, and wrote the manuscript. K.K. and J.M.S. contributed equally to this work. K.K. guided the formal analysis and statistical approach. J.M.S. verified the analysis code and provided student-level data from the Multitudes and language-based measures. M.Z. assisted with the administration of language-based measures and reviewed the manuscript. L.Y. contributed to data curation. C.T.-F. reviewed the manuscript in detail and provided substantive feedback. H.C. provided expert feedback on the results from a reading and dyslexia research perspective. F.P. and P.B. contributed to project administration and resources. J.D.Y. supervised the project, oversaw the results and findings, and reviewed and edited the manuscript. M.L.G.-T. acquired funding, contributed to project administration, and reviewed and edited the manuscript. All authors reviewed and approved the final version for submission.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.cub.2026.04.055.

Received: January 13, 2026
Revised: March 11, 2026
Accepted: April 24, 2026
Published: May 20, 2026

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
De-identified behavioral and assessment data from kindergarten and first-grade participants	This paper	OSF: https://osf.io/85h6e/files/osfstorage
Experiment code: Multi-Element Processing (MEP) tasks	This paper	GitHub: https://github.com/yeatmanlab/roar-mep; Zenodo: https://doi.org/10.5281/zenodo.19429971
Experiment code: Motion processing task	This paper	GitHub: https://github.com/yeatmanlab/roar-mp; Zenodo: https://doi.org/10.5281/zenodo.19425956
Analysis code: Visual tasks as language-agnostic early identification measures of reading challenges	This paper	GitHub: https://github.com/yeatmanlab/Visual-Tasks-as-Language-agnostic-Early-Identification-Measures-of-Reading-Challenges; Zenodo: https://doi.org/10.5281/zenodo.19429953
Experimental models: Organisms/strains		
Human participants: Kindergarten and first-grade students from California public schools	UCSF Dyslexia Center - Multitudes	UCSF IRB Approval #21-34782
Software and algorithms		
R statistical software	R core team[72]	https://www.r-project.org.
mice (Multiple Imputation by Chained Equations), R package	van Buuren Groothuis-Oudshoorn[73,74]	https://cran.r-project.org/package=mice; RRID: SCR_017538
Boruta feature selection algorithm, R package	Kursa and Rudnicki[51]	https://cran.r-project.org/package=Boruta
ROSE (Random Over-Sampling Examples), R package	Lunardon et al.[52]	https://cran.r-project.org/package=ROSE
mclust (Model-based Clustering and Latent Profile Analysis), R package	Scrucca et al.[49]	https://cran.r-project.org/package=mclust; RRID: SCR_016554
jsPsych (JavaScript library for behavioral experiments)	De Leeuw.[75]	https://www.jspsych.org; RRID: SCR_014425
ROAR (Rapid Online Assessment of Reading) platform	ROAR technical Manual[76]	https://roar.stanford.edu/technical
Other		
Woodcock-Johnson IV Tests of Achievement (WJ IV) — Letter-Word Identification, Word Attack, Spelling, Passage Comprehension, Sentence Reading Fluency subtests	Schrank et al.[77]	https://www.riverpub.com/products/wj4/index.html

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Participating school recruitment

Participants were recruited by the University of California, San Francisco (UCSF) Dyslexia Center's Multitudes project, an initiative supported by the State of California to develop a universal screener for reading challenges and dyslexia in California public schools. The sample was drawn from schools in locations and communities intended to be representative of the state in terms of race, ethnicity, socioeconomic status, and home language. Care was taken to reach remote rural, urban, and suburban communities to ensure that the sample was truly representative of the diversity of lived experiences in California. The study aimed to reduce selection bias with a passive consent process. The Institutional Review Board of UCSF determined that a process of informing parents about the study and how their children would be participating, with clear instructions on how to opt-out through communications with their school administrators, was appropriate for the research going into a universal screener. Parents of eligible kindergarten and first-grade students received an information sheet describing the project, which involved normal classroom activities and presented no more than minimal risk to participants. Children whose parents wished to opt out did not participate in study activities.

General Procedure

A team of 3-10 UCSF-trained proctors administered these assessments to kindergarten and first-grade students in each of the participating schools based on the school's convenience. Reading outcome measures were individually administered to each student. In each school, kindergarten and first-grade children were brought in batches to complete a battery of language, reading, and visual measures. All children were assigned a unique student tracking ID and were randomly grouped in batches. Thus, missing data means that either a child was absent on the day or did not wish to participate. Demographics reported by the participating school consisted of reported primary language of the student, age, English proficiency designation, grade, as well as race and ethnicity information. The student-level data on FRPM eligibility reported in this study was discontinued after the 2022-2023 school year. This data was available only for a selected set of children from the schools that provided FRPM eligibility information. Figure S7 shows the percentage of children reported as eligible compared to the eligibility percentage reported by the California Department of Education. The subset of children (reported in this study) who received the visual measures are thoroughly reported in a previous study on design and validation of these measures (in Ramamurthy et al.[38] see Table S1). Care was taken to reach remote rural, urban, and suburban communities to ensure that the sample was truly representative of the diversity of lived experiences in California. The demographic information of the children in our study is comparable to that reported by the State of California (https://www.cde.ca.gov/ds/ad/ceffingertipfacts.asp), thereby confirming the representativeness that we aimed for.

Sociodemographic Group Definitions

Children were grouped based on available data on:

a. *Free and reduced-price meal (FRPM)*: The National School Lunch Program (NSLP) provides free meals for children whose household income is less than 130% of the poverty line and reduced-price meals for income between 130%–185% of the poverty line. FRPM metrics are highly correlated with county poverty rates (0.91 - 0.95).[60] FRPM eligibility data is a student-level indicator of home income. We compared performance across students who were eligible or ineligible for FRPM. Ineligibility indexes higher socioeconomic status and was coded as 0 in our dataset.
b. *Student's primary language*: All students whose parents had indicated to the school that English is the only language spoken at home were categorized as English-only (coded as EO). Those students labeled as English learners (coded as EL) whose reported primary language was Spanish were classified as Spanish speakers. These two groups combined comprised ~90% of students in the dataset.

The correlation between FRPM eligibility and a student's primary language was 0.267 (phi coefficient for binary predictors) although we expected that these two predictors will be correlated with each other more strongly. We examined how students across different grades performed in the visual tasks and found no group differences in terms of FRPM (Figure S1) or primary language (Figure S2).

METHOD DETAILS

Battery of assessments

Standardized Reading measures

The Woodcock-Johnson IV Tests of Achievement were administered to all children at the end of the school year to assess key literacy skills through Letter-Word Identification, Word Attack, and Spelling tests. For this study, we used the letter word identification (WJ-LWI) subtest as the reading outcome for both English and Spanish-speaking children. It should be noted these reading outcome measures in English are a measure of children's classroom reading demands.

The Letter-Word Identification test evaluates basic reading skills by measuring, untimed, word recognition and decoding abilities, it requires them to read and correctly identify letters and words that range from high-frequency to those of increasing difficulty. The LWI tests are integral components of the broader WJ IV battery, designed to comprehensively evaluate single-word reading abilities and written language capabilities across various developmental stages.[72,73,78] To capture more comprehensive profiles of reading ability, we also examined composite scores from combining subtests derived from the WJ IV battery:

i. *Basic Reading Skills*: Composed of the letter-word identification and word attack subtests, this composite evaluates fundamental decoding skills, including sight word recognition, phonemic decoding, and structural analysis. It reflects a child's capacity to accurately read both familiar and novel words using a combination of visual and phonological strategies.
ii. *Broad Reading composite*: This composite integrates performance across letter-word identification, passage comprehension, and sentence reading fluency subtests. It provides a multidimensional index of reading achievement, encompassing decoding accuracy, reading fluency (rate), and comprehension. The composite score offers insight into how effectively students read connected text and extract meaning under time constraints.

iii. *Reading composite*: Comprising letter-word identification and passage comprehension, this composite encompasses decoding and reading comprehension (only administered to first graders). It captures the ability to both recognize individual words and understand meaning in sentences and brief passages, thereby offering a reliable estimate of core reading competence.
iv. *Reading Fluency*: This reflects performance in the sentence reading fluency subtest for first graders.

Across all these reading outcome measures, it is important to acknowledge that for Spanish-speaking children, performance may partially reflect English knowledge or ongoing language acquisition rather than pure decoding or automaticity. This conflation of linguistic and reading skills is a known limitation when using English-based assessments in emergent bilingual populations and underscores the need for interpreting reading scores in light of children's instructional language context and exposure.

Reading Risk
We chose the 20th percentile of the within-sample distribution of the WJ-LWI as the cut-off for risk classification (this was based on a previous study on the same dataset[50]).

Visual-processing measures
Included rapid visual processing of multiple elements of letters and pseudo letters and global motion processing ability. All visual measures were administered using a touchscreen Chromebook and all instructions were through an engaging audio-visual illustration of the task.

Multi-element Processing letters (MEP-L) & Multi-element Processing pseudo-letters (MEP-P)
For details on task description, trial structure, design, and validation please see Ramamurthy et al.[38] The assessment was gamified to make it more engaging and fun for younger children. The task simply involved a string of elements that were briefly presented for 240ms. After the elements disappeared a blue line appears below one of the element positions and participants were required to identify the element that was presented above the blue line from a set of six choices. The game sets children on a mission to help a lost whale in the sea, by playing this game of elements that is the secret map to the lost treasures and friends.

Motion
This experiment was built using JS psych and launched online on the ROAR app platform.[76] The stimuli for the motion task were adapted from a previous study.[31] In this single-interval forced-choice experiment, on each trial random dots with varying proportions of signal and noise were presented. The assessment was gamified to make it more engaging and fun for younger children. The task simply involves participants judging the global direction of motion of the entire dot field. The game sets children on a mission to help Beatrice the bee, in finding where the colony of bees is headed in a dense forest. If the colony of bees is headed to the apple tree they must click on the apple tree on the left or the lemon tree on the right.

Task understandability
For the visual measures, we wanted to distinguish task understandability from task performance. Blinded from the demographics of the participating students, we first cleaned all the data from the MEP and Motion tasks based on a predetermined rubric for performance.

MEP task filtering criterion
For the MEP tasks only children who attempted and engaged with at least 24 trials were considered for further analysis (a total of 1457 attempted these tasks). In addition, those participants whose raw scores were below 4 correct out of the 24 attempted trials had ability estimates that were overestimated by the IRT model. These participants are highlighted in green in Figures S3A and S3B and were excluded from further analysis. Figure S3C shows the reading outcome distribution for the excluded children 115 (MEP-L) and 135 (MEP-P) confirming that we were not categorically excluding poor readers.

Motion task filtering criterion
This task had a non-response contingent trial succession, meaning trials are a fixed length time and the game automatically will progress to the next trial. Therefore, it is critical to define a performance cut-off that would inform us about the child's task understandability. All participants who surpassed a performance cut-off of > 57% in the highest signal coherence conditions of 96% were filtered and included (a total of 934 children attempted this task). This cut-off in performance was to verify that children understood the task and performed above chance in at least the easiest trials. This criterion excluded around 220 participants. The excluded participants reading outcome distributions were normal and we weren't categorically eliminating poor readers (see Figure S3D).

Language-based measures
A pilot version of the *Reach Every Reader (RER) digital* assessment battery[77] with individual tasks that were widely used in screening was administered.[79] These tasks tapped into constructs of phonological awareness (deletion, blending), decoding (letter naming, word reading, non-word reading), phonological working memory (digit span, non-word repetition), rapid object naming, and oral language (expressive vocabulary, sentence repetition). We selected the Woodcock-Johnson Letter-Word Identification (WJ LWI) subtest as our reading outcome measure, as it assesses word reading skills—a core component of decoding, therefore the additional decoding measures were not included in the language model.

Administration and data collection of RER language-based measures
All language measures were administered on iPads with headphones. The administration involved a proctor-facing iPad and a student-facing iPad. Students were asked to wear headphones for instructions and prompts. All instructions were very engaging and child-friendly audio-visual illustrations of the task. Data were collected by UCSF proctors working individually with each child, ensuring the accuracy and efficiency of administration and data collection. Most measures were designed as computer adaptive

testing and the platform returns ability estimates after each child completes the task. For measures that are not computer-adaptive like rapid automatic naming of objects, the platform recorded raw scores.

Procedure

Phonological awareness was measured using the blending (BLE) and deletion (DEL) tasks. For the blending task, students join two phonemes or words to form a single word and remove one phoneme or word from a given word for the deletion task. Decoding skills were assessed using Word Reading (WRE) and Non-Word Reading (NRE), where students read the word or nonsense word presented on screen. Language and vocabulary skills were assessed with expressive vocabulary (EVO) and sentence repetition (SRT) tasks. EVO involved naming the visually presented object correctly and in SRT students repeated a set of sentences. Letter Naming context (LNC) and rapid automatic naming of Objects (RAN) skills were both time-sensitive measures and were scored based on the total number of correct responses in 45 seconds. In the RAN task, children were presented with a grid of 5x5 objects and a practice to name all the objects. The task is to name the objects in the 5x5 grid as quickly as possible. Similarly, for the LNC task, children were presented with random upper and lowercase letters and were asked to read out the grid of 5x5 letters as quickly as possible. Phonological working memory was assessed using the digit span (DGS) task and non-word repetition (NWR) tasks. In the DGS task, students repeated two-digit number sequences that incrementally went up to six digits. The task was discontinued after three consecutive incorrect responses. In the NWR task, students repeated a list of nonsense words. Reliability of language-based measures: The empirical reliability of the language measures used in this study ranged from.61 to.97. Most of the criterion-referenced reliability for the language measures ranged from.54 to.78. These numbers indicate moderate to high consistency across the majority of measures (see Appendix B Table B1 in a previous study[50]).

The timeline of how all these measures were administered is illustrated in Figure 7.

QUANTIFICATION AND STATISTICAL ANALYSIS

Data Imputations

To address missing values and preserve statistical power across a wide range of predictors, we performed multiple imputation using the *mice* package in R.[74] Imputations were conducted separately for kindergarten and first grade cohorts before each analysis. Imputations were applied to distinct subtests of variables based on their domain: the domains were: i. language-based measures (deletion, bending, non-word repetition, sentence repetition, digit span, rapid automatic naming, evocative vocabulary); ii. rapid visual processing measures (MEP-L and MEP-P); iii. Motion (is its own domain); iv. reading outcome at the end of the same school year (WJ: LWI, WA, ORF, RF, SRF) and v. reading outcome measures at the end of the next school year (WJ: LWI, WA, ORF, RF, SRF, Spelling). Before imputations the number of missing data for each grade were as follows: K (n =157): blending = 23; deletion = 22; Non-word repetition = 22; sentence repetition = 32; digit span = 38, rapid automatic naming = 27; MEP-L = 0; MEP-P =34, Motion = 53, WJ-LWI_concurrent year = 0; WJ-LWI_ year-later = 13; WJ- BasicReadingSkills (year-later) = 14; WJ- BroadReading (year-later) = 64; WJ-reading (year-later) = 13. For first grade(G1) (n=248): deletion = 58; Non-word repetition = 59; sentence repetition = 62; digit span = 70, rapid automatic naming = 60; MEP-L = 0; MEP-P =23, Motion = 66, WJ-LWI_concurrent year = 1; WJ-LWI_ year-later = 3; WJ- BasicReadingSkills (year-later) = 3; WJ- BroadReading (year-later) =20; WJ-reading (year-later) = 3; WJ-reading fluency = 20.

Within each subset, missing values were imputed using predictive mean matching (PMM) and a fixed random seed to ensure reproducibility. After imputation, the subsets were merged using a common student identifier to generate a single, fully imputed dataset per grade. The imputed kindergarten and first grade datasets were then combined for all downstream analyses. Importantly, imputation was performed prior to the calculation of outcome-based classifications (e.g., risk status) to avoid introducing bias. This approach enabled the integration of diverse behavioral and cognitive data while preserving the natural variance structure and ensuring the reproducibility of the analysis pipeline. The reason why many children received one measure or the other is often due to logistical reasons. Since data were collected in schools time and human resources were limited and proctors had to sometimes choose. For example, if there is limited time they always administer MEP-L (because we expect performance to be correlated with MEP-P as reported previously[38]). This pattern is consistent with missing at random (MAR), the assumption under which multiple imputation via predictive mean matching produces unbiased estimates that we used.

Feature importance plots

To identify which language-based and visual predictors most reliably distinguished children at risk for reading difficulties, we employed the Boruta feature selection algorithm,[51] a wrapper-based method built on random forests that iteratively compares the importance of actual features against permuted “shadow” features. This approach allows for the identification of all relevant features without relying on arbitrary thresholds or assumptions about linearity.

Data preparation

We selected a comprehensive set of predictor variables based on theoretical relevance to early reading development. These included measures of rapid visual processing (MEP–L; MEP–P; Motion), language skills (sentence repetition and expressive vocabulary), phonological awareness (deletion, blending), rapid automatized naming of objects (RAN), and phonological working memory (digit span and non-word repetition). Each variable was z-scored prior to analysis for each grade. The binary reading risk outcome was derived based on WJ at the end of first grade.

Two datasets were constructed: 1. An original (imbalanced) dataset with class imbalances of those at risk; 2. A synthetically balanced dataset generated using the ROSE (Random Over-Sampling Examples) package in R,[52] which synthesizes new data points by bootstrapping and interpolating between existing minority class observations to counteract the effects of class imbalance.

Feature selection was conducted separately on both the imbalanced and balanced datasets using the Boruta function in R with 100 random forest iterations and 100 trees per forest. The outcome variable was reading risk status, and all language-based and visual predictors were included as candidate features. Feature importance was assessed using Z-score normalized mean decrease in accuracy values, and predictors were ranked based on their median importance scores across iterations. A custom post-processing function was applied to clean and extract the Boruta output. Importance values were visualized using boxplots ordered by median importance as shown in Figures 4 and S4.

Latent profile analysis

To identify subgroups of children with distinct profiles based on performance, we conducted a model-based latent profile analysis (LPA) using the mclust package in R.[49] This approach fits gaussian finite mixture models under various constraints to the covariance matrices. Preferred models were selected using the Bayesian Information Criterion (BIC). All continuous input variables were standardized (z-scored) prior to modeling for each grade separately.

The LPA was first run on kindergarten children (n = 157). We chose the EEI model—which assumes equal volume and shape for each group with diagonal covariance matrices across aligned to the coordinate axes (Figures S5A and S5B). The EEI parameterization assumes the grouping variables to be uncorrelated and offers a balanced compromise between model complexity and parsimony, particularly suitable for relatively small samples where overparameterization can result in unstable solutions. Among all models evaluated, the EEI model with five latent profiles achieved the best BIC value (BIC = −3878.19) compared to the 4-profile solution (BIC = −3878.55; ΔBIC = 0.36). Given the statistical equivalence of these two solutions, final model selection was additionally informed by interpretive criteria[80]: the core profiles — including the visual strength, visual challenge, and above-average performing groups — were preserved intact across both solutions, while the 5-profile solution (Figures S5C and S5D) further differentiated average and below average subgroups distinguished primarily by performance in rapid automatized naming and digit span.

Classification reliability of the 5-profile solution was strong in the full imputed sample (entropy = 0.895, exceeding the recommended 0.80 threshold,[81] indicating that the five profiles are well-separated and that children were assigned to their profiles with high confidence. Profile-level average posterior probabilities ranged from 0.891 to 0.993, including for the two smallest subgroups, confirming that small group size did not compromise classification certainty. To separately assess whether the profile structure was sensitive to imputation, we reran the LPA on a complete-case subsample excluding Motion — the variable with the highest missingness (34%) — yielding n = 90 children with fully observed data. The 5-profile solution was fully replicated, with entropy increasing to 0.945 and all profile means preserved in direction across every variable, confirming that the profile structure is not an artifact of imputation. Nevertheless, we encourage replication in larger samples using the methods detailed here before these subgroup distinctions are broadly generalized.

The five subgroups identified by this solution, in our imputed sample (n=157), reflected a range of distinct performance patterns, defined here to facilitate replication. These included an above-average performing group with scores $\geq$ +0.25 SD across all tasks, an average-performing group with scores near the sample mean, and a below-average performing group with scores $\leq$ – 0.25 SD. The visual strength group exhibited scores in the MEPs $\geq$ +0.25 SD and scores on deletion < -0.10 and the visual challenge group exhibited scores in deletion $\geq$ +0.10 SD and scores on MEPs < -0.25 SD. These empirically derived profiles were subsequently applied to the cohort of first graders to assess the generalizability of the classification framework across samples. All five profiles were represented with adequate sample sizes in Grade 1 — Above-average Performers (n = 37), Average Performers (n = 108), Below-average Performers (n = 35), Visual Strength (n = 42), and Visual Challenge (n = 19) — with all profiles exceeding the 5% non-spurious classification threshold[80] and four of five exceeding the n = 30–60 generalizability range,[82] providing preliminary cross-sample evidence for the stability of the kindergarten-derived profiles.

FRPM and Primary Language Group Analyses

Figure S1A shows how children from the FRPM Eligible and Ineligible groups perform across all tasks. We observed no significant difference in group performance between the two FRPM groups for the visual measures, but a large effect size (Cohen's d) is observed for the reading outcome measure. The number of participants with student-level data was about half the sample size for each measure because new regulation makes it more difficult for schools to share student-level data, so we also used the openly available school-level data on percentage of students with FRPM eligibility. The school-level FRPM eligibility data was available for the school year 2022-2023 from the California Department of Education (https://dq.cde.ca.gov/dataquest/). We fit a linear model to the school-level median scores regressed on these measures with percentage of FRPM eligibility reported for each school and grades as predictors. As shown in Figure S1B, the coefficients of FRPM as predictor was close to zero for the visual measures compared to other measures, most evidently for the reading outcome measures as substantiated in Figure 5.

Since English learners in California specifically are predominantly Spanish speakers (https://www.cde.ca.gov/ds/ad/cefelfacts.asp), we next compared children whose reported primary spoken-language at home was English with those who reported speaking Spanish. English learners who were Figure S2A, shows how the effect size of the difference between the English and Spanish groups compares across visual, reading, and language measures. Visual-processing measures are unbiased to differences in the primary language of the participants for kindergarten and the non-alphabetic visual tasks are unbiased to differences in the primary language of the participants in first grade. Note that all the visual tasks were administered in English, but children were screened for task understandability (see STAR Methods).

Like Figure S1B, we fit a linear model to the school-level median scores regressed on these measures with groups based on the primary language of participants from each school and grade as predictors. As shown in Figure S2B, the coefficients of primary language as predictor were close to zero for the model predicting visual-processing measures compared to other measures, most evidently for the reading outcome measure (WJ-LWI) as substantiated in Figure 6 as boxplots.

Regression Models: Visual Measures, FRPM, and Primary Language

Regression models were fit on the available FRPM and primary language data to corroborate with the correlation plots presented in Figures 5G–5I and 6G–6I. Model summaries are presented in Tables S1 and S3.

The first set of models regresses the reading outcome on all three visual measures, separately, and their interactions with FRPM eligibility,

$$Y = \beta_0 + \beta_1 \mathrm{MEP} - \mathrm{L} + \beta_2 \mathrm{MEP} - \mathrm{P} + \beta_3 \mathrm{Motion} + \varepsilon_i. \quad \text{(Equation 1)}$$

$$Y = \beta_0 + \beta_1 \mathrm{MEP} - \mathrm{P} + \beta_2 \mathrm{Motion} + \varepsilon_i \quad \text{(Equation 2)}$$

$$Y = \beta_0 + \beta_1 \mathrm{MEP} - \mathrm{L} + \beta_2 \mathrm{MEP} - \mathrm{P} + \beta_3 \mathrm{Motion} + \beta_4 \mathrm{FRPM} + \beta_5 \mathrm{MEP} - \mathrm{L} \times \mathrm{FRPM} + \beta_6 \mathrm{MEP} - \mathrm{P} \times \mathrm{FRPM} + \beta_7 \mathrm{Motion} \times \mathrm{FRPM} + \varepsilon_i \quad \text{(Equation 3)}$$

while the second model is the same specification, replacing FRPM eligibility with an indicator of primary language:

$$\mathbf{Y} = \beta_0 + \beta_1 \mathrm{MEP} - \mathrm{L} + \beta_2 \mathrm{MEP} - \mathrm{P} + \beta_3 \mathrm{Motion} + \beta_4 \mathrm{FRPM} + \beta_5 \mathrm{MEP} - \mathrm{L} \times \mathrm{FRPM} + \beta_6 \mathrm{MEP} - \mathrm{P} \times \mathrm{FRPM} + \beta_7 \mathrm{Motion} \times \mathrm{FRPM} + \varepsilon_i \quad \text{(Equation 4)}$$